

MUSIX

Intelligent Playlist Optimizer

Where Data Structures Meet Music.

DSA Semester Project | C Programming

BCA – Computer Applications



What is MUSIX?

A smart playlist system powered by practical Data Structures.

MUSIX helps organize songs and perform playlist operations efficiently.



MANAGE

Add and organize songs in a structured playlist.



DISCOVER

Search songs and access playlist information quickly.



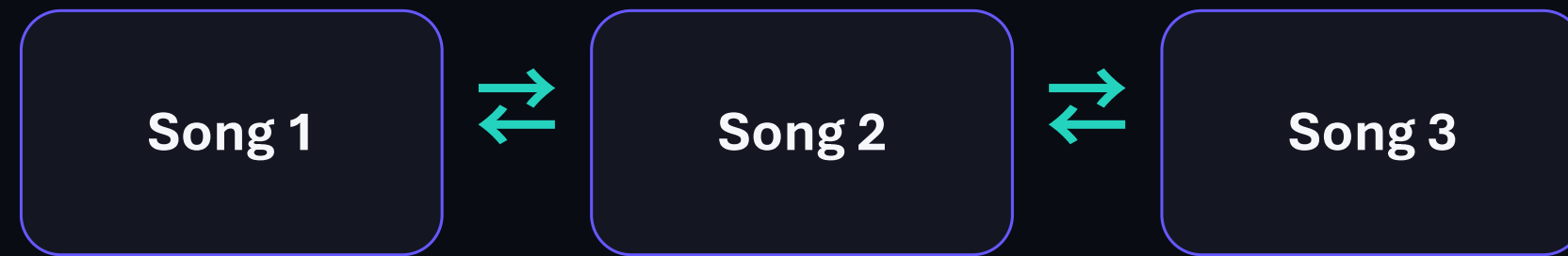
OPTIMIZE

Use queue, history and time-based playlist logic.

Core Data Structures

Each structure has a specific role inside MUSIX.

DOUBLY LINKED LIST



Main playlist storage

→ QUEUE

Stores upcoming songs
in play order.
[FIFO]

↶ STACK

Keeps recently played
songs.
[LIFO]

DSA is not just theory — MUSIX uses each structure for a real task.

Key Features

Six features that make MUSIX more than a basic song list.



PLAYLIST

Organize songs efficiently.



SEARCH

Find a song quickly.



SMART QUEUE

Manage upcoming songs.



HISTORY

Track recently played songs.



FIT MY TIME

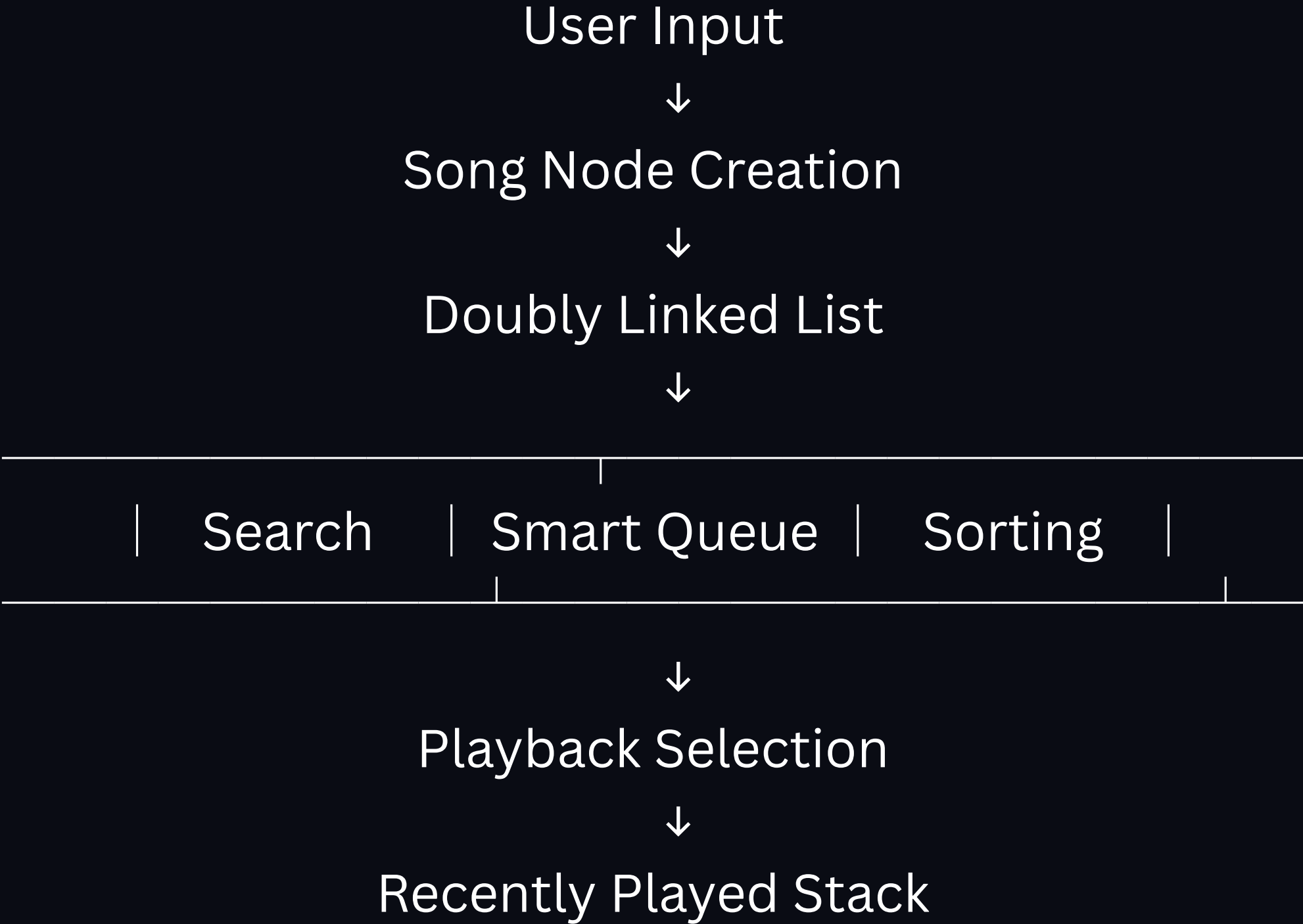
Build a playlist for available time.



NO REPETITION

Avoid unnecessary repeated songs.

Technical Architecture



Why MUSIX?

A compact project with clear DSA implementation and practical purpose.

MUSIX converts core DSA concepts into a practical playlist management system.



PRACTICAL

Solves real playlist-management tasks.



TECHNICAL

Demonstrates Linked List, Queue and Stack.



EXTENSIBLE

Can grow into a GUI, database or recommendation system.

Simple Music • Smart Data Structures

THANK YOU